Editing facets over REST

Applies to `glineRestApi` v2.4 and later. Correct `units` handling requires v2.41.3 or later.

This document covers one thing: writing a component's **facets** — units, precision, range — from a single `mod` operation, and the two traps that make it fail.

What facets are

`facets` is the slot that tells Niagara **how to display** a value: its unit, how many decimal places, its allowed range. It is not the value itself, and it is not the component's *device* facets (the ones on the proxy extension). It is the plain `facets` slot present on any point.

```
{"OP":"mod","PATH":"/Drivers/SnmpNetwork1/PDU_AP8858_180/points/Phase1_Voltage",
 "N":"facets",
 "V":"units=volt;precision=1;min=0;max=300"}
```

Response:

```
{"OP":"mod","path":"/Drivers/SnmpNetwork1/PDU_AP8858_180/points/Phase1_Voltag
e","N":"facets","status":"ok"}
```

Read it back:

```
{"OP":"read","PATH":"/Drivers/SnmpNetwork1/PDU_AP8858_180/points/Phase1_Voltag
e","N":"facets"}
```

```
{"OP":"read","path":"/Drivers/SnmpNetwork1/PDU_AP8858_180/points/Phase1_Voltag
e","N":"facets",
 "value":"units=V,precision=1.00,min=0.00,max=300.00","status":"ok"}
```

Note the readback renders a **unit symbol** (`V`), not the name you wrote (`volt`).
That is the signature of a correctly-stored unit — see trap 1.

The facet string

`V` is a `key=value` list. **Fields are separated by a semicolon (`;`).**

Key	Type written	Accepted values
`units`	`BUnit`	a Niagara unit name, e.g. `volt`, `ampere`, `watt`, `volt ampere`, `kilowatt hour`, `hertz`, `percent`, `celsius`, `day`, `hour`, `second`
`precision`	`BDouble`	decimal places, e.g. `0`, `1`, `2`
`min`	`BDouble`	numeric
`max`	`BDouble`	numeric
`range` / `enumrange`	`BEnumRange`	`A,B,C` or `0:A,1:B,2:C`

Example — voltage with one decimal, 0…300 V:

```
units=volt;precision=1;min=0;max=300
```

Example — power factor, no unit, two decimals, 0…1:

```
precision=2;min=0;max=1
```

Clearing a unit: write `units=null` (or omit `units` entirely — the whole facet value is replaced, not merged).

Trap 1 — `,` is not a separator (v2.4 → v2.41.2)

Only `;` splits fields. A comma is **not** a separator, so this:

```
units=volt,precision=1,min=0,max=300
```

is parsed as **one** field — key `units`, value `volt,precision=1,min=0,max=300` — and the entire string is stored as the unit. Every field after the first is lost, and the unit is junk.

Always use `;`:

```
units=volt;precision=1;min=0;max=300
```

Trap 2 — storing a unit as text (v2.4 → v2.41.2)

`units` is typed **`javax.baja.units.BUnit`** — an object — not a string. Through v2.41.2 the facet writer stored non-numeric facet values as `BString`, so `units` went in as text. The readback looked plausible (`units=V`), but the slot held a `BString`.

The failure only appears when something *formats a number with that unit* — which is exactly what Workbench does when it renders the point:

```
java.lang.ClassCastException: javax.baja.sys.BString cannot be cast to javax.baja.
units.BUnit
    at javax.baja.sys.BDouble.toString(BDouble.java:393)
    at javax.baja.status.BStatusNumeric.valueToString(BStatusNumeric.java:148)
    at com.tridium.wiresheet.PropertyBarGlyph.updateValueString(PropertyBarGlyph.j
ava:60)
```

From v2.41.3, `units` is resolved to a real `BUnit` via `BUnit.getUnit(name)`. A name that does not resolve is rejected with a clear error instead of being stored:

```
{"OP":"mod","error":"Unknown unit 'bogus'. Use a Niagara unit name, for example: v
olt, ampere, watt, kilowatt, volt ampere, kilowatt hour, hertz, percent, celsius,
day, hour, second."}
```

How to tell a good unit from a bad one: read `facets` back. A real unit renders its **symbol** (`V`, `A`, `W`, `VA`, `kW-hr`, `s`). Text renders whatever you typed (`volt`).

Clearing a unit written before v2.41.3: re-write the facet string with the fields you want and **no `units`** (or `units=null`). The facet value is replaced wholesale, so the old text unit is dropped.

```
precision=1;min=0;max=300
```

Common unit names

Names verified against the station's unit database:

Quantity	Name to write	Renders as

Voltage	`volt`	`V`
Current	`ampere`	`A`
Active power	`watt`	`W`
Active power (large)	`kilowatt`	`kW`
Apparent power	`volt ampere`	`VA`
Apparent power (large)	`kilovolt ampere`	`kVA`
Energy	`kilowatt hour`	`kW-hr`
Frequency	`hertz`	`Hz`
Ratio	`percent`	`%`
Temperature	`celsius`, `fahrenheit`, `kelvin`	`°C`, `°F`, `K`
Duration	`second`, `millisecond`, `minute`, `hour`, `day`	`s`, `ms`, `min`, `hr`, `day`
Data	`byte`, `bit`	`byte`, `bit`

Two-word names are written with a **space**, not a hyphen: `volt ampere`, `kilowatt hour` — not `volt-ampere` / `kilowatt-hour`. A name that is not in the database is rejected.

Verifying the result

A facet write is only correct if the value *renders* with it. Check all three:

1. **Readback shows a symbol** — `units=V`, not `units=volt`:

   ```
   {"OP":"read","PATH":".../points/Phase1_Voltage","N":"facets"}
   {"OP":"read","value":"units=V,precision=1.00,min=0.00,max=300.00","status":"o
   k"}
   ```

2. **The point's value reads without fault** — a bad unit faults on formatting:

   ```
   {"OP":"read","PATH":".../points/Phase1_Voltage","N":"out"}
   {"OP":"read","value":"225.00 {ok}","status":"ok"}
   ```

3. **It renders in Workbench** — open the point on a wire sheet. A `ClassCastException` stack trace in the Workbench console means a text unit is still in the slot.

Worked example — APC AP8858 PDU, 14 points

All written with `;`, all read back as symbols.

Point	Facet string	Readback
Phase1_Voltage	`units=volt;precision=1;min=0;max=300`	`units=V,precision=1.00,min=0.00,max=300.00`
Phase1_Current	`units=ampere;precision=1;min=0;max=20`	`units=A,…`
Phase1_PeakCurrent	`units=ampere;precision=1;min=0;max=20`	`units=A,…`
Phase1_Power	`units=watt;precision=0;min=0;max=4000`	`units=W,…`
Phase1_ApparentPower	`units=voltampere;precision=0;min=0;max=4000`	`units=VA,…`
Phase1_PowerFactor	`precision=2;min=0;max=1`	`precision=2.00,min=0.00,max=1.00`
Phase1_LoadState	`precision=0;min=0;max=4`	`precision=0.00,…`
PDU_Power	`units=watt;precision=0;min=0;max=4000`	`units=W,…`
PDU_PeakPower	`units=watt;precision=0;min=0;max=4000`	`units=W,…`
PDU_Energy	`units=kilowatthour;precision=1;min=0`	`units=kW-hr,precision=1.00,min=0.00`
PDU_ApparentPower	`units=voltampere;precision=0;min=0;max=4000`	`units=VA,…`
PDU_PowerFactor	`precision=2;min=0;max=1`	`precision=2.00,min=0.00,max=1.00`
PDU_LoadState	`precision=0;min=0;max=4`	`precision=0.00,…`
SysUpTime	`units=second;precision=0;min=0`	`units=s,…`

Points with real values then read as, for example, `225.00 {ok}` (V), `5.00 {ok}` (A), `32557.00 {ok}` (kW-hr).

Summary

Rule	Detail
Field separator	`;` — a comma is not a separator
`units`	a Niagara unit name; resolved to `BUnit` from v2.41.3
Two-word units	space, not hyphen (`volt ampere`)
Clearing a unit	omit `units`, or `units=null`; the facet value is replaced wholesale
Verified by	readback renders a **symbol**, and `out` reads without fault
Pre-v2.41.3	units were stored as text and crashed Workbench on render — upgrade, or leave units unset

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